Shadowseed — Manual

Version 0.12.4 · Research Preview

PolyForm Noncommercial License 1.0.0 · github.com/E-AI-MODEL/shadowseed-pro

Contents

Part 1 — Getting started

1. What Shadowseed is
2. System requirements
3. Installing Shadowseed
4. Setting up a local model with Ollama
5. Your first conversation

Part 2 — Using Shadowseed

6. The screen
7. Seeds and their statuses
8. Authority modes
9. Adding verified evidence
10. Contradictions
11. When a seed influences an answer
12. Using OpenAI
13. Your data and privacy
14. Limitations
15. Troubleshooting

Part 3 — Testing

16. Testing Shadowseed

Part 4 — Research

17. The Shadow Seed Learning model
18. Showing that SSL had an effect
19. Studies and evidence

Part 5 — Development

20. Working from source
21. Architecture
22. Command-line reference
23. Web API reference
24. Builds and releases

Part 6 — Security and operations

25. Security model
26. Running Shadowseed reliably
27. Configuration reference

Appendices

A. Glossary (English–Dutch)
B. Frequently asked questions
C. About this document

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This manual covers the downloadable Shadowseed web application and the Python package it is built from. It is written for five groups of readers. Everyone should read Part 1. After that, go to the part that fits your role.

You are…	Read
A **user** who wants to chat with Shadowseed	Parts 1 and 2
A **tester** who reports findings	Parts 1, 2 and 3
A **researcher** who studies Shadow Seed Learning	Parts 1, 2 and 4
A **developer** who builds on or changes Shadowseed	Parts 1, 4 and 5
A **specialist** in security, privacy or operations	Parts 1, 5 and 6

The glossary at the end translates every term into Dutch. A Dutch edition of this manual is available as `shadowseed-handleiding-nl.md`.

> **Status.** Shadowseed is a *Research Preview*. It works and is tested, but it is not certified for decisions in healthcare, finance, law, employment, education or other high-impact areas. Chapter 14 lists its limitations.

Part 1 — Getting started

1. What Shadowseed is

Shadowseed is a chat application with an extra memory layer called **Shadow Seed Learning (SSL)**. You talk to a language model as usual. Next to the conversation, Shadowseed watches each answer for points that are still open: an assumption that was not checked, a number that was not given, a case that was not covered. It stores such a point as a **seed**.

A seed starts with no influence at all. It can gain influence only through the **Validation Gate**, and in the default mode the Gate wants a source that *you* have checked. Once a seed has passed the Gate, Shadowseed may bring it back into a later answer — but only when it is relevant to the question at that moment.

Three ideas carry the whole design:

- **Observe broadly, grant authority selectively, retrieve narrowly.** Many seeds are noticed; few are allowed to steer; only the relevant ones are used.
- **Authority comes from evidence, not from repetition.** A seed that comes back often is interesting, but repetition alone does not make it true.
- **Every decision is visible.** Each seed has a timeline of events, and every Gate decision is recorded.

Shadowseed does not retrain the model. The model stays the same; the memory around it grows. Two installations with the same model can therefore develop different memories.

What Shadowseed does **not** claim:

- that seeds are true;
- that answers become better (this is the subject of ongoing research);
- that a source you mark as checked is actually correct — that is your responsibility.

2. System requirements

Item	Requirement
Operating system	Windows 10/11 (64-bit), macOS on Apple silicon (M1 or later), Linux x86_64
Disk space	About 200 MB for Shadowseed, plus space for models (a 7B model takes about 4.7 GB)
Memory	8 GB RAM minimum for a 7B model; 16 GB recommended
Ollama	Version 0.11.10 or later, for real conversations
Browser	A current Chrome, Edge, Firefox or Safari

Item	Requirement
Network	Not needed for local use, except to download the program and the models

Speed. On a computer without a suitable graphics card, a 7B model needs about two to three minutes per turn. A GPU or a smaller model shortens this considerably. The offline demo needs no model and responds instantly.

3. Installing Shadowseed

3.1 Download

Go to the release page of the project on GitHub and download the archive for your system:

System	File
Windows	`shadowseed-web-0.12.4-windows-amd64.zip`
macOS (Apple silicon)	`shadowseed-web-0.12.4-darwin-arm64.zip`
Linux	`shadowseed-web-0.12.4-linux-x86_64.tar.gz`

Also download `SHA256SUMS` if you want to verify the file.

3.2 Verify the download (recommended)

The checksum proves that your file is exactly the one that was published.

- **Windows (PowerShell):** `Get-FileHash .\shadowseed-web-0.12.4-windows-amd64.zip`
- **macOS:** `shasum -a 256 shadowseed-web-0.12.4-darwin-arm64.zip`
- **Linux:** `sha256sum shadowseed-web-0.12.4-linux-x86_64.tar.gz`

Compare the result with the matching line in `SHA256SUMS`. They must be identical.

3.3 Unpack and start

Windows. Right-click the ZIP and choose *Extract All*. Open the extracted folder, then the `Shadowseed` folder, and start `Shadowseed.exe`. Windows may show a SmartScreen warning, because the program is not signed with a Microsoft certificate. Choose *More info* and then *Run anyway*.

macOS. Double-click the ZIP to unpack it. The application is signed for integrity but not notarized by Apple, so macOS blocks the first start of a downloaded copy. Right-click `Shadowseed` and choose *Open*, then confirm. If macOS still refuses, run once in Terminal: `xattr -dr com.apple.quarantine /path/to/Shadowseed.app`.

Linux. Unpack with `tar -xzf shadowseed-web-0.12.4-linux-x86_64.tar.gz` and start the `Shadowseed` program in the extracted folder.

After starting, Shadowseed opens your web browser at an address such as `http://127.0.0.1:8765/`. That address exists only on your own computer; no one else can reach it. If port 8765 is taken, Shadowseed picks another free port and opens that instead.

Keep the window or terminal of Shadowseed open while you work. Closing it stops the program. Your conversations are kept.

3.4 Where your data is stored

Shadowseed keeps everything in one folder in your home directory:

System	Folder
Windows	`C:\Users\<name>\.shadowseed`
macOS	`/Users/<name>/.shadowseed`
Linux	`/home/<name>/.shadowseed`

Next to it, a folder `.shadowseed-integrity` holds the protected key and anchor of the audit ledger. Do not copy, move or delete these folders by hand; use the commands in chapter 13: `workspace delete` removes both folders, and a backup goes back only through `workspace restore`. Chapter 13 also explains what is stored.

4. Setting up a local model with Ollama

The offline demo shows how Shadowseed works without a model. For real conversations Shadowseed uses **Ollama**, which runs language models on your own computer.

1. Install Ollama from ollama.com (version 0.11.10 or later) and start it.
2. Download a chat model. For example, in a terminal: `ollama pull qwen2.5:7b-instruct`
3. Download the embedding model that Shadowseed needs to compare texts: `ollama pull embeddinggemma`
4. Check that both are present: `ollama list`

Any chat model that Ollama can run will work. Larger models give better answers and find more useful seeds, but need more memory and time.

Which model? In our measurements, a 7B model on a normal computer finds a seed in about one turn in eight, and about a quarter of those seeds are useful. Larger hosted models find far more useful seeds (51 of 60 in a sample). If you mainly want to see the mechanism, the offline demo or a small model is enough. If you want useful seeds, use the largest model your computer can run.

5. Your first conversation

5.1 Start with the offline demo

The offline demo uses no model. Its answers are simple echoes and its seeds are marked `[FIXTURE]`. It is meant to show the controls, not to give good answers.

1. In the left panel under **New conversation**, give the conversation a title.
2. Leave **Model** on *Offline demo (no model)*.
3. Leave **Authority mode** on *Controlled*.
4. Click **Start conversation**.
5. Type a message and press **Enter**.

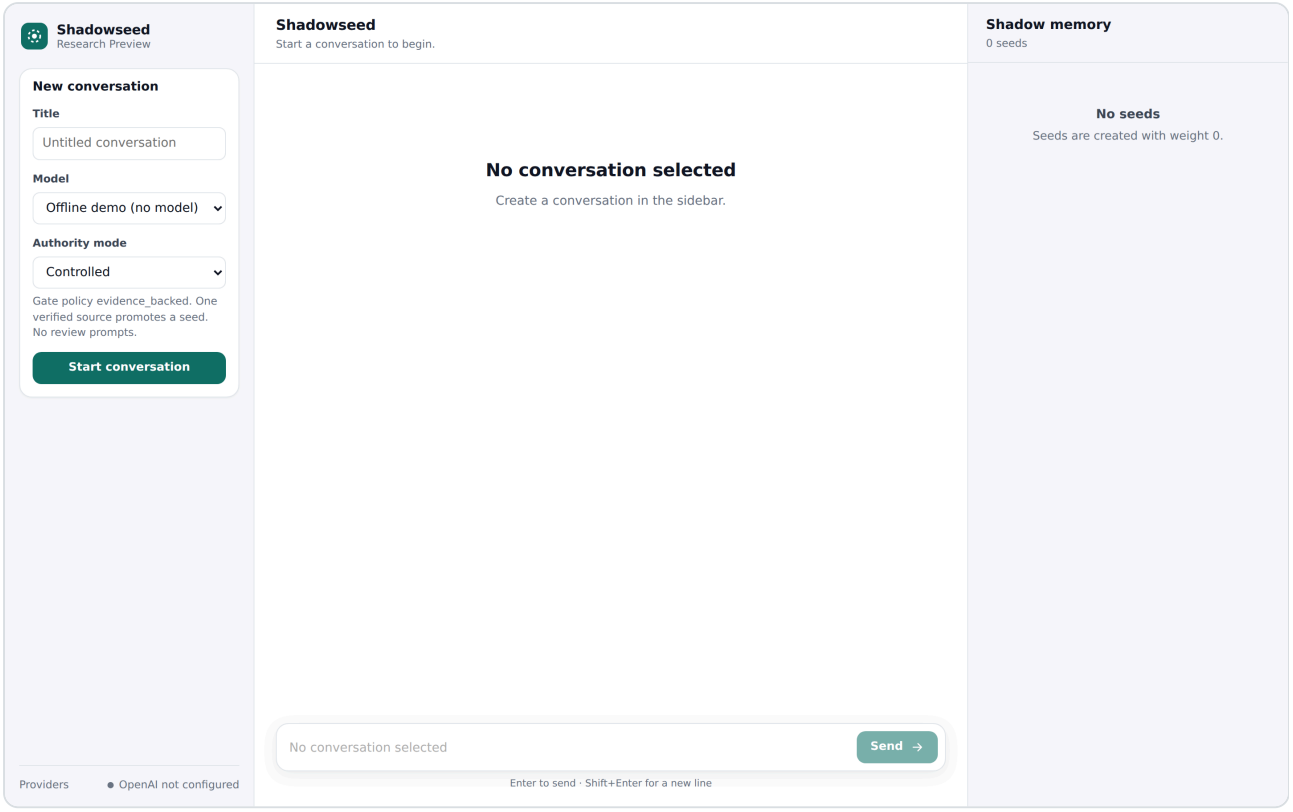


Figure 1 — The start screen. Left: new conversation. Middle: the chat. Right: Shadow memory.

5.2 A real conversation with Ollama

1. Click **New conversation**.
2. Under **Model**, choose *Ollama (local)*.
3. Under **Ollama model**, enter the exact name of an installed model, for example `qwen2.5:7b-instruct`. Use the name exactly as `ollama list` shows it.
4. Choose an authority mode (chapter 8) and click **Start conversation**.

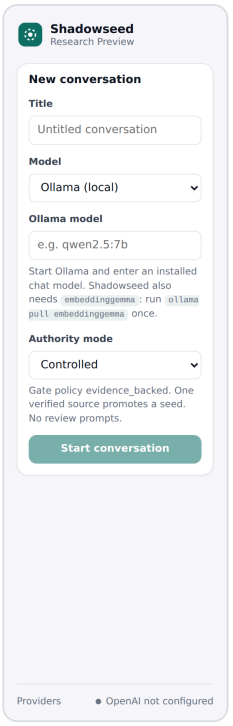


Figure 2 — Creating a conversation with a local Ollama model.

If Ollama is not running, or `embeddinggemma` is missing, Shadowseed shows what to do and does not create the conversation. Fix the problem and try again.

Ask a real question and continue the conversation for a while. Seeds appear in the right panel as Shadowseed finds open points. The longer the conversation, the more chance that a point comes back.

Part 2 — Using Shadowseed

6. The screen

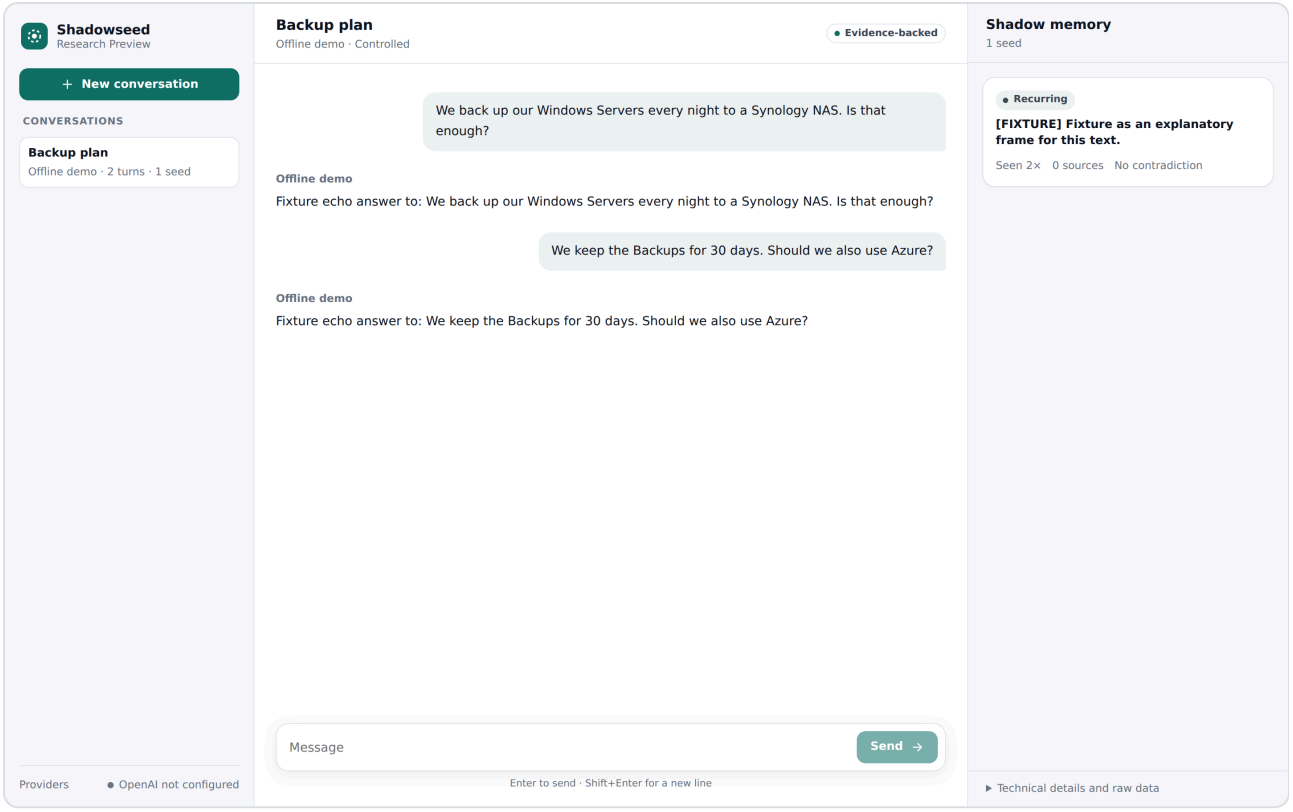


Figure 3 — A conversation with one seed in Shadow memory.

The screen has three panels.

Left — Conversations. Your conversations, newest first, with the model, the number of turns and the number of seeds. **New conversation** opens the form. At the bottom, **Providers** shows whether OpenAI is available (chapter 12).

Middle — The chat. The header shows the title, the model, the authority mode and the Gate policy badge (for example *Evidence-backed*). Type below and press **Enter** to send; **Shift+Enter** starts a new line. While the model is working you see *Waiting for response…*.

Right — Shadow memory. All seeds of this conversation. Each card shows the status, the text, how often it was seen, how many verified sources support it, how often it was used, and whether a contradiction is open. Click a card to open the seed. At the bottom, **Technical details and raw data** shows the Gate policy, the authority profile, the behavior digest and the full session data as JSON.

On a phone or narrow window, the side panels fold away. Use the menu button for conversations and the **Shadow** button for Shadow memory.

7. Seeds and their statuses

A seed is a short, traceable note about a point that may be open. It is a *hypothesis*, not a fact and not an instruction.

Every seed has a status. The status tells you what the seed may do.

Status	Meaning	Can it influence an answer?
Found	Noticed once.	No
Recurring	Noticed again in a later turn.	No
Supported	Has at least one verified source, but the Gate has not given it weight.	No
Validated	Has weight from the Gate, but is not (or no longer) promoted.	No
Promoted	Passed the Gate. Eligible for use when relevant.	Yes, when relevant
Used	Promoted and actually used in at least one answer.	Yes, when relevant
Blocked	Has an open contradiction.	No
Dormant	Not seen for a while; its trace has faded. Can wake up when the topic returns.	No
Expired	Faded completely. Kept for the record.	No

A *Recurring* seed in **Assisted** mode can carry the mark *Action required*: it keeps coming back without a source. Shadowseed then asks you to look at it instead of granting influence on its own.

Opening a seed

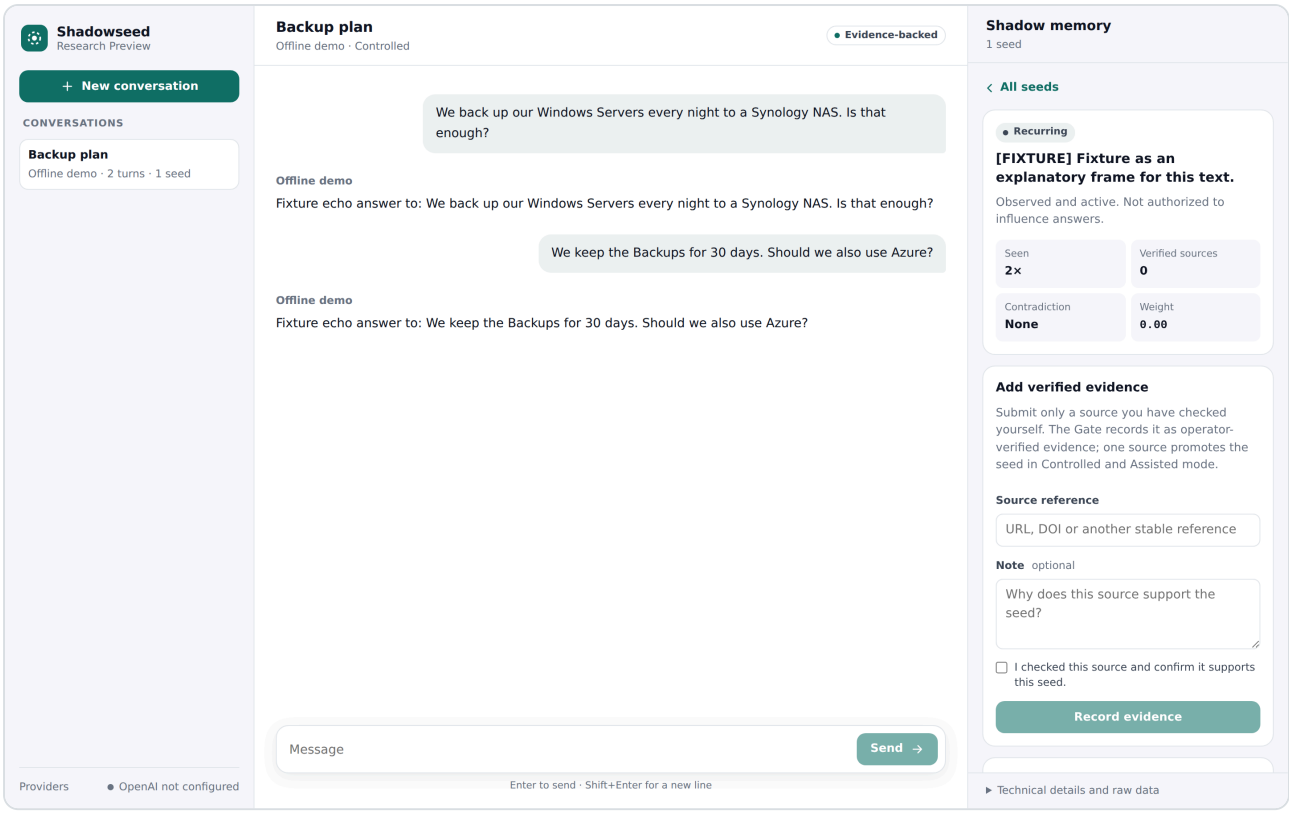


Figure 4 — An opened seed with its counters and the evidence form.

The detail view shows:

- **Seen** — in how many turns the seed was observed;
- **Verified sources** — how many distinct checked sources support it;
- **Contradiction** — whether a contradiction is open;
- **Weight** — its steering power, between 0 and 1. New seeds start at 0;
- **Used in answers** — how often it influenced an answer;
- **Trace** — a fading measure of recent presence;
- forms for evidence and contradiction, the **Timeline**, and **Raw data**.

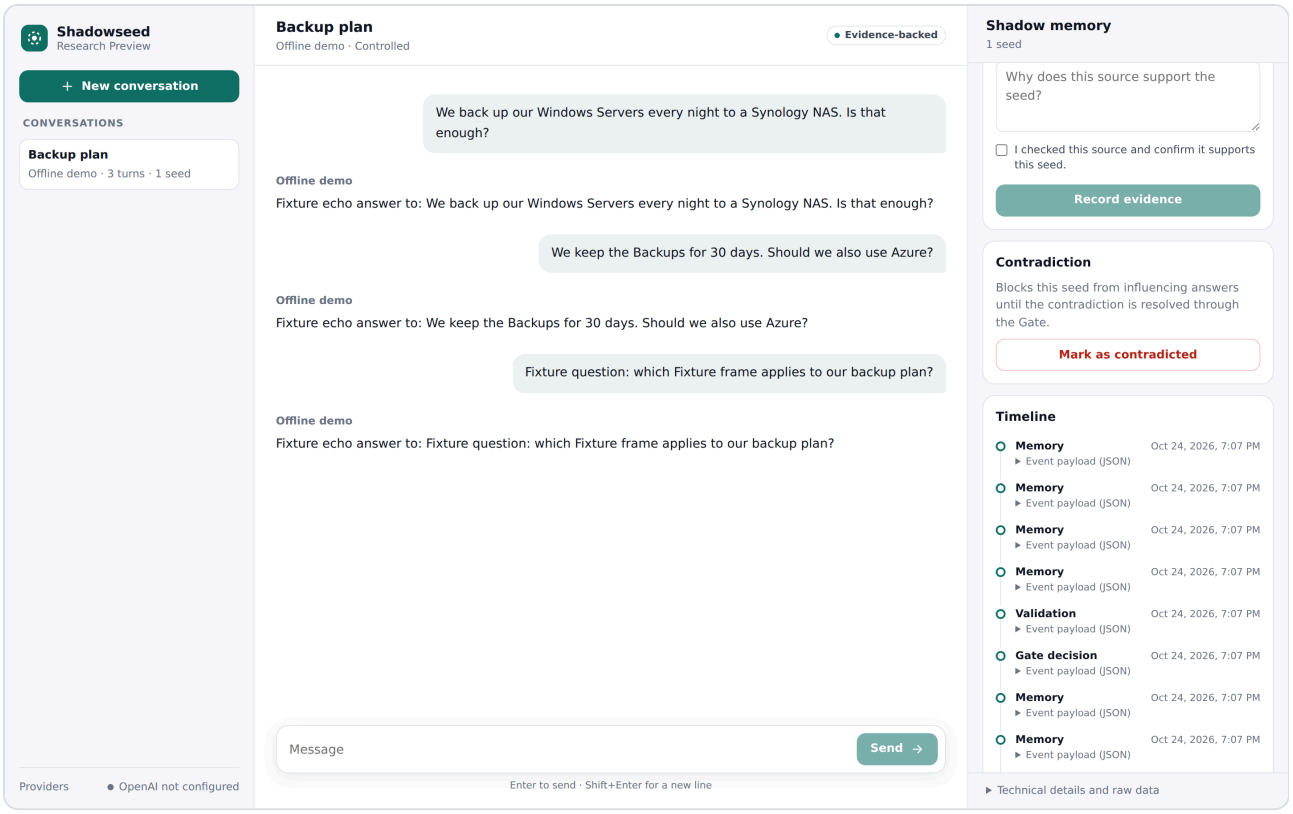


Figure 5 — The timeline. Every memory event, validation and Gate decision is listed with its time. Open an event to see the full record.

How seeds recur

When a new note might be the same open point as an earlier seed, Shadowseed asks the model one short question: is this the same open question, or only the same topic? Only a "same question" answer counts as recurrence. If the model cannot answer, Shadowseed falls back to text similarity with a conservative threshold. Recurrence is recorded but never gives authority on its own in Controlled and Assisted mode.

8. Authority modes

You choose the authority mode when you create a conversation. It decides how a seed can gain influence.

Mode	Gate policy	How a seed gains authority	Use it for
Controlled (default)	Evidence-backed	Only through a source you have checked. One verified source promotes the seed.	Normal use
Assisted	Evidence-backed	As Controlled, and Shadowseed asks for your attention when a seed keeps recurring without a source.	When you want to be prompted
Autonomous	Exploratory	Recurrence counts as support; no external source is needed.	Research only

In the technical details, Controlled appears as authority profile `strict`.

Autonomous mode lets the system grant authority without your involvement. Use it to study the mechanism, not for answers you rely on.

9. Adding verified evidence

A seed in Controlled or Assisted mode can only become *Promoted* through a source that you have checked yourself.

1. Open the seed.
2. Under **Add verified evidence**, enter a **Source reference**: a URL, a DOI or another stable reference.
3. Optionally add a **Note** explaining why the source supports the seed.
4. Tick **I checked this source and confirm it supports this seed.**
5. Click **Record evidence**.

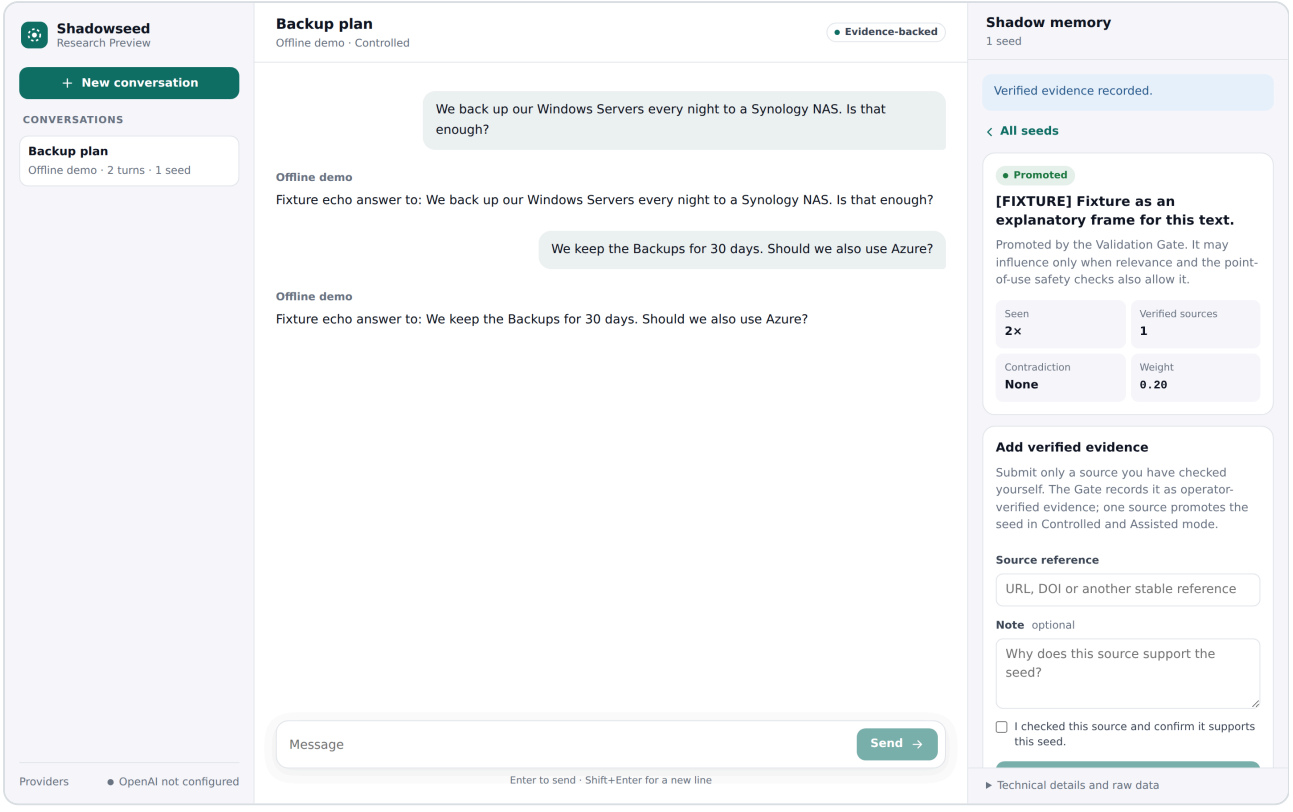


Figure 6 — After one verified source the seed is Promoted, with weight 0.20.

Rules that protect the Gate:

- **Only check what you have read.** The confirmation is recorded in the audit ledger as your statement.
- **One source counts once.** Entering the same reference again does not add authority, even through another route.
- **Model output is not evidence.** Do not cite an answer of the model or a summary it wrote.

- **Recurrence is not evidence.** A seed that came back often still needs a source.

10. Contradictions

If you find that a seed is wrong, misleading or out of date:

1. Open the seed.
2. Under **Contradiction**, click **Mark as contradicted**.

The seed is now **Blocked** and cannot influence any answer.

To lift the block, enter a **Basis for resolution** — why the contradiction no longer applies — and click **Resolve contradiction**. Resolving only removes the block. The seed does **not** get its authority back automatically; it must pass the Gate again with a new verified source.

11. When a seed influences an answer

A seed influences an answer only if all of these hold at the moment you ask:

1. it is **Promoted**, with weight above zero;
2. its promotion is backed by a current Gate decision;
3. it has no open contradiction;
4. it is **relevant** to your current question;
5. the **point-of-use check** allows it for this turn.

So *Promoted* does not mean *used*. A promoted seed about backup retention stays silent when you ask about something else. That is intended.

When a seed is used, its status becomes **Used** and the counter **Used in answers** goes up. The seed reaches the model as bounded context marked as untrusted; the model is not told to treat it as fact.

One more rule: when a seed was used in an answer, new seeds found in that same answer are set aside for that turn, so Shadowseed does not learn from its own influence.

12. Using OpenAI

The downloadable application works with local models only. OpenAI models are available when Shadowseed is installed as a Python package with the `openai` extra (chapter 20). In that case:

- enter a key under **Providers → Temporary API key**. The key stays in memory only; it is not saved and disappears when Shadowseed stops;
- when you create an OpenAI conversation, you must agree that prompts, shadow context and embeddings are sent to OpenAI;
- you confirm this again for each message you send.

13. Your data and privacy

What is stored. The folder `.shadowseed` in your home directory contains a local database with your conversations, the model answers, seeds, Gate decisions, evidence references and notes, and the audit ledger. Its protected integrity key lives in the neighbouring folder `.shadowseed-integrity`. Nothing is sent anywhere unless you use OpenAI.

What not to type. Do not paste passwords, API keys, access tokens or other secrets into conversations, evidence references or notes. Use made-up or anonymized material when real personal data is not needed.

Backups and deletion. The web application has no export button yet. With the Python package installed you can use:

- `shadowseed workspace info` — show where the data is and how much there is;
- `shadowseed workspace backup --output <file>` — make a consistent copy of the database. Always give a file **outside** `.shadowseed`: without `--output` the copy goes to `.shadowseed/exports`, and deleting the workspace deletes it too;
- `shadowseed workspace restore <file>` — restore a copy;
- `shadowseed workspace delete --yes` — delete the complete workspace;
- `shadowseed export-workbench-report <session>` — export one conversation as a verified report (contains the full content);
- `shadowseed export-support-bundle <session>` — export a privacy-minimized support bundle (no conversation or seed text).

The `<session>` id is shown under **Technical details and raw data → Session**.

A backup contains the same content as the original and must be protected in the same way. Deleting the workspace removes everything inside `.shadowseed`, including `.shadowseed/exports`; copies you saved elsewhere are kept.

14. Limitations

- **Few seeds with small local models.** With a 7B model, a seed appears in about one turn in eight and about a quarter of the seeds are useful.
- **Speed on CPU.** Two to three minutes per turn with a 7B model on a computer without a suitable GPU.
- **No proof of better answers.** Whether SSL improves answers is the subject of ongoing studies. A pilot study exists; a full result does not yet.
- **Seeds can be wrong.** A seed is a hypothesis. A promoted seed is supported by your source, not proven true.
- **Single user, single computer.** Shadowseed is not a multi-user or network service.
- **Not signed by Apple or Microsoft.** Your system warns on first start.
- **Interface in English.** The application itself is currently English only.
- **No comparison button.** The web application does not yet offer the "same question without SSL" comparison; it is available through the web API (chapter 18).

15. Troubleshooting

Problem	Likely cause	What to do
The browser does not open	Blocked by the system or the default browser is missing	Open the address shown in the Shadowseed window yourself
macOS: "cannot be opened"	Quarantine on a downloaded app	Right-click → Open, or run the `xattr` command from 3.3
Windows: SmartScreen warning	Program not signed	More info → Run anyway
"Ollama not reachable"	Ollama is not running	Start Ollama and try again
Message about `embeddinggemma`	Embedding model missing or Ollama too old	`ollama pull embeddinggemma`; update Ollama to 0.11.10 or later
"Model not found"	Model name differs from the installed name	Use the exact name from `ollama list`
A turn takes minutes	Model runs on CPU	Expected for 7B on CPU; use a smaller model or a GPU
No seeds appear	Small models often find nothing	Continue the conversation; try a larger model
Provider needed badge	The model of this conversation is not available	Start Ollama or configure OpenAI

The doctor

The Python package contains a diagnostic tool:

```
shadowseed doctor
```

It checks Python, the workspace, the database, the audit ledger, permissions, disk space, Ollama, the embedding setup and the optional providers, and tells you how to fix what it finds. Typical output:

```
python                  OK    3.11.15
workspace               OK    /home/you/.shadowseed
production_integrity    OK    verified ledger head sequence 7
ollama                  WARN  not detected at http://127.0.0.1:11434
                              Start `ollama serve`, pull a model, and rerun

`shadowseed doctor`.
Workbench can start.
```

To use the doctor with the downloaded application, install the Python package from the same release (see 20.1) and run `shadowseed doctor`; it checks the same `.shadowseed` folder.

Part 3 — Testing

16. Testing Shadowseed

16.1 What to look at

- Can you download, start and use the application without any other software?
- Is it clear which model is active and which authority mode applies?
- Does it feel like a normal chat before you look at the seeds?
- Are seed statuses and explanations understandable?
- Is it clear that evidence must be checked by you, and that recurrence is not evidence?
- Are error messages actionable when Ollama or a model is missing?
- Can you close and reopen a conversation without losing anything?

16.2 How to test

1. Start with the offline demo. It proves the mechanics, not the quality.
2. Then use Ollama with a real model and longer conversations (eight turns or more), because seeds and recurrence need time.
3. Note the model, the authority mode and what you did.

16.3 Reporting a problem

Include what you tried, the model and settings, the smallest sequence of steps that shows the problem, what you expected and what happened, and whether a seed was used in that turn. Do not include personal data or secrets.

Behavior that seems to let anyone change a seed's weight or promotion directly, outside the Gate, is a serious defect. Report it with priority.

16.4 Support bundles

With the Python package, `shadowseed export-support-bundle <session>` produces a privacy-minimized support bundle. It keeps structure and counts but leaves out conversation and seed text. Study coordinators combine bundles with `scripts/aggregate_support_bundles.py`. Do not edit a bundle by hand; its integrity check will fail.

Part 4 — Research

17. The Shadow Seed Learning model

17.1 Lifecycle

Observe → Seed → Recur / gather evidence → Validate → Promote → Relevance check → Point-of-use influence. Each step is separate, and only the Gate changes authority.

- **Detection** reads the question, the draft answer and a bounded window of earlier turns, and proposes up to five atomic candidate gaps. The prompt is `detector_current_pair` v0.6.
- **Intake** admits a candidate as a weightless seed if it is a single question, or a single English statement that is not a bare list of categories.
- **Recurrence** is recognized by the session model (ADR-012), with embedding similarity as a floor (0.3) and as fallback.
- **Trace** records presence and decays with a half-life; seeds become dormant and later expire. A dormant seed can be reactivated when its topic returns.
- **Weight** is steering power. New seeds start at 0. Only the Gate raises it.

17.2 Gate policies

Policy	Accepts
`evidence_backed`	Verified external support only. In Controlled and Assisted live sessions one source promotes (threshold 0.2, ADR-011).
`exploratory`	Qualifying recurrence *or* verified external support.
`legacy_evidence_required`	Historical thresholds for recurrence, trace and evidence; kept for compatibility.

Verified support needs a non-empty `source_ref`, which is the identity of the evidence: the same reference never counts twice. Contradictions are explicit records with an id, reason, status and resolution basis; a seed with an open contradiction is blocked. Resolution unblocks but does not restore weight.

17.3 Point of use

Before a seed may influence generation, `AgentSafetyContract.decide_and_record` checks positive weight, promoted status, no blocking contradiction, and a live Gate event for the seed's current authority version. Allowed and denied decisions are both recorded, and each allowed decision is linked to the Gate event that authorized it. Replay can detect a stale authorization.

17.4 Two kinds of memory

Under `evidence_backed`, recurrence is observation: a *trace*. Under `exploratory`, recurrence may become a support signal after three occurrences. The product default is `evidence_backed`.

18. Showing that SSL had an effect

A different wording between two answers is not evidence of SSL influence, because two generations differ anyway. Attribute an effect only when:

1. the seed was promoted;
2. it was selected as relevant for the turn;
3. the point-of-use decision allowed it;
4. it actually surfaced, as recorded in the turn report and the seed timeline.

For a paired comparison, a turn can be sent with `compare_without_ssl: true` through the web API (chapter 23). The control uses the same model, the same visible history and the same message, receives no seed context, and does not change any state. The web application stores the comparison in **Technical details → Last turn comparison**; it does not yet have a button for it.

19. Studies and evidence

The repository contains preregistered study designs and their results:

- **Seed calibration** (`docs/research/seed-calibration.md`): usefulness of detected seeds and the reliability of recurrence recognition.
- **Detector calibration** (`docs/research/detector-calibration.md`): two blind-rated runs on Qwen 2.5 7B showing that prompt changes alter the number of seeds but not their usefulness on that model.
- **Evidence efficacy** (`docs/research/evidence-efficacy.md`): does an authorized seed improve answers against SSL-off and a self-critique control? Study 1 is a pilot.

Principles that apply to every study:

- fix the design and decision rule before the run;
- rate blind, and record the labels before decoding the key;
- record the exact model digest, the source commit and the prompt versions;
- treat model output as observation, never as evidence.

Research tooling lives in the separate `research/` package and in `scripts/`. Results are committed under `benchmarks/` with their provenance.

Part 5 — Development

20. Working from source

20.1 Installing the package

From a release, without Git:

```
python -m pip install shadowseed-0.12.4-py3-none-any.whl
```

From the repository:

```
git clone https://github.com/E-AI-MODEL/shadowseed-pro.git
cd shadowseed-pro
python -m pip install -e ".[test]"
```

Optional extras: `models` (Hugging Face and Sentence Transformers), `openai`, `workbench` (legacy Gradio interface), `vector`, `build`, `standalone-build`, `security`, and `dev` for everything. Python 3.10 or later is required.

20.2 Tests and checks

```
python -m pytest -q          # full suite
ruff check src tests scripts # lint
uv lock --check              # lockfile matches pyproject
```

The web client is a Next.js application in `apps/web`. Build it and copy the static output into the Python package with `scripts/sync_web_assets.py`.

20.3 Rules for changes

- The Validation Gate is the only path that may raise authority. Do not add another.
- Authority fields (`weight`, `status`, `evidence_count`, `contradiction_score`) cannot be assigned directly; tests enforce this.
- Changes to the authority model need an ADR in `docs/architecture/adr/`.
- Every change goes through a pull request with green CI.

21. Architecture

Part	Location	Role
Core	`src/shadowseed/` (`manager.py`, `lifecycle.py`, `chat.py`)	Seeds, lifecycle, chat session

Part	Location	Role
Gate	`src/shadowseed/gate/`	The canonical Validation Gate and its ledger
Detection	`src/shadowseed/detection/`	Detector prompts and parsers
Intake and recurrence	`intake.py`, `recurrence_*.py`	Admission and recurrence
Storage	`src/shadowseed/storage/`	SQLite workspace, integrity ledger
Application	`src/shadowseed/application/`	Workspace, limits, health checks
Adapters	`src/shadowseed/adapters/`	Ollama, OpenAI, Hugging Face, embeddings
Web API	`src/shadowseed_webapi/`	Loopback HTTP server and launcher
Web client	`apps/web/`	The browser interface
Legacy Workbench	`src/shadowseed/workbench/`	Gradio interface (opt-in)
Research	`research/`	Benchmarks and studies

Design records: `docs/architecture/adr/ADR-001` to `ADR-012`. Start with ADR-001 (the Gate as sole authority), ADR-006 (production-local boundary), ADR-011 (single-source promotion) and ADR-012 (model recurrence recognition).

22. Command-line reference

Command	Purpose
`shadowseed-web [--port N] [--workspace DIR] [--no-browser]`	Start the web application (default port 8765, loopback only)
`shadowseed-web --self-test --workspace DIR`	Run the built-in self-test
`shadowseed chat --backend fixture|ollama|openai`	Chat in the terminal
`shadowseed doctor [--json] [--workspace DIR]`	Diagnose the installation
`shadowseed init [--workspace DIR]`	Create a workspace
`shadowseed workspace info|backup|restore|delete`	Manage the workspace

Command	Purpose
`shadowseed-workbench [--legacy-gradio]`	Start the desktop launcher: the web client by default, the legacy Gradio interface with `--legacy-gradio`
`shadowseed export-workbench-report <session>`	Export one conversation as a verified, content-bearing report ZIP
`shadowseed export-support-bundle <session>`	Export a privacy-minimized support ZIP

Run any command with `--help` for all options. `docs/usage/cli.md` documents the research and benchmark commands.

23. Web API reference

The server listens on `127.0.0.1` only. All routes start with `/api/v1`.

Method	Route	Purpose
GET	`/health`	Health check
GET	`/providers`	Provider status
POST	`/providers/openai/credential`	Set a temporary OpenAI key
POST	`/providers/openai/credential/clear`	Remove it
GET	`/sessions`	List conversations
POST	`/sessions`	Create a conversation (`title`, `backend`, `model_id`, `authority_mode`, `external_confirmed`)
GET	`/sessions/{id}`	Read a conversation
POST	`/sessions/{id}/turns`	Send a message (`question`, `request_id`, optional `compare_without_ssl`, `external_confirmed`)
GET	`/sessions/{id}/seeds/{seed}`	Read a seed with its timeline
POST	`/sessions/{id}/seeds/{seed}/evidence`	Add verified evidence (`source_ref`, `note`, `operator_verified: true`, `request_id`)
POST	`/sessions/{id}/seeds/{seed}/contradictions`	Mark as contradicted

Method	Route	Purpose
POST	`/sessions/{id}/seeds/{seed}/contradictions/resolve`	Resolve a contradiction (basis required)

Sending a turn and the three seed actions (evidence, contradiction, resolution) require a `request_id`: replaying the same id is safe, and a reused id with different content is refused. Creating a conversation and setting or clearing the OpenAI key have no `request_id`; repeating them repeats the action. Requests from a foreign browser origin or with a non-loopback `Host` header are refused.

24. Builds and releases

- **CI** runs the tests on Python 3.10 and 3.12, production-local checks on Windows, macOS and Linux, a strict audit of the shipped dependencies, and frozen builds for all three platforms on every pull request.
- **Release Workbench** (manual) publishes the version in `pyproject.toml` from the exact `main` commit, after CI on that commit is green. It builds the three downloads, `SHA256SUMS`, `PROVENANCE.json`, an SBOM, wheel and sdist.
- **Production Release Assurance** runs automatically on the published tag and verifies attestations, checksums and provenance.
- Release notes live in `docs/workbench/release-<version>.md`.

Part 6 — Security and operations

25. Security model

Claimed for the local profile:

- the server is reachable only through loopback (`127.0.0.1`); the `Host` header is validated against DNS rebinding;
- authority-changing actions are attributed to a trusted local actor;
- a client flag alone cannot create verified authority;
- the Gate and the point-of-use contract are the only path to authority and influence;
- tampering with the authority history, or rolling back to an older state, is detectable against protected integrity material;
- operational logs are content-minimized;
- releases are tied to an exact source commit with checksums and provenance.

Not claimed:

- protection against a compromised operating system or administrator;
- multi-user isolation or security on a hostile network;
- the factual truth of evidence that an operator marks as checked;
- universal protection against prompt injection;
- better answers in general;
- secure erasure beyond what the file system provides.

The full threat model is in `docs/architecture/production-threat-model.md`.

26. Running Shadowseed reliably

- **Limits.** Oversized input is refused before anything is stored. Model calls have a 120-second timeout and no automatic retries. A failure never changes the stored state.
- **Permissions.** On macOS and Linux, the workspace is readable by its owner only. On Windows, the user account protects it.
- **Integrity.** An audit ledger with protected integrity material records authority events. Never "repair" a workspace by deleting its integrity material; restore a verified backup instead.
- **Backups.** Use `shadowseed workspace backup --output <file>` with a file outside the workspace; the default location inside `.shadowseed/exports` is removed with the workspace. Keep backups as safe as the original.
- **Containers.** `Dockerfile.workbench` is for development. Publish the port on loopback only: `-p 127.0.0.1:7860:7860`.
- **Diagnostics.** `shadowseed doctor` separates usable state, integrity problems and missing dependencies.

Operational procedures: `docs/operations/production-local-recovery.md` and `docs/operations/production-local-release.md`.

27. Configuration reference

Setting	Default	Meaning
`--port`	8765	Preferred port; another free port is used if it is taken
`--workspace`	`~/.shadowseed`	Data folder
`OLLAMA_HOST`	`http://localhost:11434`	Address of the Ollama server
`OPENAI_API_KEY`	not set	OpenAI key for the Python package; the web application uses the temporary key instead
Authority mode	Controlled	Set per conversation
Embedding model (Ollama)	`embeddinggemma`	Needed for real conversations

Appendices

A. Glossary (English–Dutch)

English	Dutch	Meaning
Shadow Seed Learning (SSL)	Shadow Seed Learning (SSL)	The memory layer around the model
Seed	Seed (zaadje)	A traceable note about a possibly open point
Shadow memory	Schaduwgeheugen	All seeds and their history
Validation Gate	Validatiepoort (Gate)	The only place where authority changes
Authority	Zeggenschap	The right of a seed to influence answers
Weight	Gewicht	Steering power of a seed, 0 to 1
Trace	Spoor	Fading measure of recent presence
Recurrence	Terugkeer	A seed seen again in a later turn
Verified evidence	Geverifieerd bewijs	A source the operator checked
Contradiction	Tegenspraak	A recorded objection that blocks a seed
Promoted	Gepromoveerd	Passed the Gate; eligible for use
Used	Gebruikt	Actually used in an answer
Point of use	Moment van gebruik	The final check before influence
Authority mode	Zeggenschapsmodus	Controlled, Assisted or Autonomous
Gate policy	Gate-beleid	`evidence_backed` or `exploratory`
Workspace	Werkmap	The `.shadowseed` data folder
Audit ledger	Auditlogboek	Protected record of authority events
Offline demo (fixture)	Offline demo	Mode without a model

B. Frequently asked questions

Does Shadowseed send my conversations anywhere? No, unless you choose OpenAI. With Ollama everything stays on your computer.

Why do I see so few seeds? Small models find few open points. Longer conversations and larger models help.

I added a source, but the seed is not used. Why? A promoted seed is only used when it is relevant to the question you ask. Ask about the topic of the seed.

Can I set a seed's weight myself? No. Only the Gate changes authority. That is a deliberate safeguard.

Does a seed that recurs often become promoted? Not in Controlled or Assisted mode. Recurrence is recorded but needs a verified source to become authority. In Autonomous mode recurrence can count.

Is a promoted seed true? Not necessarily. It is supported by the source you checked. Check sources carefully.

May I use Shadowseed commercially? The license is PolyForm Noncommercial 1.0.0. Commercial use needs separate permission.

C. About this document

Applies to	Shadowseed 0.12.4
Editions	English (`shadowseed-manual-en.md`), Dutch (`shadowseed-handleiding-nl.md`); PDF versions in the same folder
Screenshots	Offline demo in the web application of version 0.12.4
Source	`docs/manual/` in the repository
License	PolyForm Noncommercial License 1.0.0

Corrections and suggestions are welcome as an issue in the repository.